

Srinivasa Ramanujan

Born: 22 Dec 1887 in Erode, Tamil Nadu state, India
Died: 26 April 1920 in Madras, Tamil Nadu state, India



Ramanujan was one of India's greatest mathematical geniuses. He made substantial contributions to the analytical theory of numbers and worked on elliptic functions, continued fractions, and infinite series.

**Srinivasa Ramanujan (Dec. 22, 1887 --
April 26, 1920)**

K. Srinivasa Rao

The Institute of Mathematical Sciences, Madras-600 113.

Srinivasa Ramanujan (1887-1920) hailed as an all-time great mathematician, like Euler, Gauss or Jacobi, for his *natural genius*, has left behind 4000 original theorems, despite his lack of formal education and a short life-span. In his formative years, after having failed in his F.A. (First examination in Arts) class at College, he ran from pillar to post in search of a benefactor. It is during this period, 1903-1914, he kept a record of the final results of his original research work in the form of entries in two large-sized Note Books. These were the ones which he showed to Dewan Bahadur Ramachandra Rao (Collector of Nellore), V. Ramaswamy Iyer (Founder of Indian Mathematical Society), R. Narayana Iyer (Treasurer of IMS and Manager, Madras Port Trust), and to several others to convince them of his abilities as a Mathematician. The orchestrated efforts of his admirers, culminated in the encouragement he received from Prof. G.H. Hardy of Trinity College, Cambridge, whose warm response to the historic letter of Ramanujan which contained about 100 theorems, resulted in inducing the Madras University, to its lasting credit, to rise to the occasion thrice - in offering him the first research scholarship of the University in May 1913 ; then in offering him a scholarship of 250 pounds a year for five years with 100 pounds for passage by ship and for initial outfit to go to England in 1914 ; and finally, by granting Ramanujan 250 pounds a year as an allowance for 5 years commencing from April 1919 soon after his triumphant return from Cambridge ``with a scientific standing and reputation such as no Indian has enjoyed before".

Ramanujan was awarded in 1916 the B.A. Degree by research of the Cambridge University. He was elected a Fellow of the Royal Society of London in Feb. 1918 being a ``Research

student in Mathematics Distinguished as a pure mathematician particularly for his investigations in elliptic functions and the theory of numbers" and he was elected to a Trinity College Fellowship, in Oct. 1918 (- a prize fellowship worth 250 pounds a year for six years with no duties or condition, which he was not destined to avail of). The "Collected Papers of Ramanujan" was edited by Profs. G.H.Hardy, P.V. Seshu Aiyar and B.M. Wilson and first published by Cambridge University Press in 1927 (later by Chelsea, 1962 ; and by Narosa, 1987), seven years after his death. His 'Lost' Notebook found in the estate of Prof. G.N. Watson in the spring of 1976 by Prof. George Andrews of Pennsylvania State University, and its facsimile edition was brought out by Narosa Publishing House in 1987, on the occasion of Ramanujan's birth centenary. His bust was commissioned by Professors R. Askey, S. Chandrasekhar, G.E. Andrews, Bruce C. Berndt ('the gang of four'!) and 'more than one hundred mathematicians and scientists who contributed money for the bust' sculpted by Paul Granlund in 1984 and another was commissioned for the Ramanujan Institute of the University of Madras, by Mr. Masilamani in 1994. His original Note Books have been edited in a series of five volumes by Bruce C. Berndt ("Ramanujan Note Books", Springer, Parts I to V, 1985 onwards), who devoted his attention to each and every one of the three to four thousand theorems. Robert Kanigel recently wrote a delightfully readable biography entitled : "The Man who knew Infinity : a life of the Genius Ramanujan" (Scribners 1991; Rupa & Co. 1993). Truly, the life of Ramanujan in the words of C.P. Snow: "is an admirable story and one which showers credit on nearly everyone".

During his five year stay in Cambridge, which unfortunately overlapped with the first World War years, he published 21 papers, five of which were in collaboration with Prof. G.H. Hardy and these as well as his earlier publications before he set sail to England are all contained in the "Collected Papers of Srinivasa Ramanujan", referred earlier. It is important to note that though Ramanujan took his "Note Books" with him he had no time to delve deep into them. The 600 formulae he jotted down on loose sheets of paper during the one year he was in India, after his meritorious stay at Cambridge, are the contents of the 'Lost' Note Book found by Andrews in 1976. He was ailing throughout that one year after his return from England (March 1919 - April 26, 1920). The last and only letter he wrote to Hardy, from India, after his return, in Jan. 1920, four months before his demise, contained no news about his declining health but only information about his latest work : "I discovered very interesting functions recently which I call 'Mock' theta-functions. Unlike the 'False' theta-functions (studied partially by Prof. Rogers in his interesting paper) they enter into mathematics as beautifully as ordinary theta-functions. I am sending you with this letter some examples ... ". The following observation of Richard Askey is noteworthy: "Try to imagine the quality of Ramanujan's mind, one which drove him to work unceasingly while deathly ill, and one great enough to grow deeper while his body became weaker. I stand in awe of his accomplishments; understanding is beyond me. We would admire any mathematician whose life's work was half of what Ramanujan found in the last year of his life while he was dying".

As for his place in the world of Mathematics, we quote Bruce C Berndt: "Paul Erdos has passed on to us Hardy's personal ratings of mathematicians. Suppose that we rate mathematicians on the basis of pure talent on a scale from 0 to 100, Hardy gave himself a score of 25, Littlewood 30, Hilbert 80 and Ramanujan 100". G.H.Hardy, in 1923, edited Chapter XII of Ramanujan's second Notebook on Hypergeometric series which contained 47 main theorems, many of them followed by a number of corollaries and particular cases. This work had taken him so many weeks that he felt that if he were to edit the entire Notebooks "it will take the whole of my lifetime. I cannot do my own work. This would not be proper." He urged Indian authorities and G.N.Watson and B.M. Wilson to edit the Notebooks. Watson

and Wilson divided the task of editing the Notebooks - Chapters 2 to 13 were to be edited by Wilson and Chapters 14 to 21 by Watson. Unfortunately, the premature death of Wilson, in 1935, at the age of 38, aborted this effort. In 1957, with monetary assistance from Sir Dadabai Naoroji Trust, at the instance of Professors Homi J Bhabha and K. Chandrasekaran, the Tata Institute of Fundamental Research published a facsimile edition of the Notebooks of Ramanujan in two volumes, with just an introductory para about them. The formidable task of truly editing the Notebooks was taken up in right earnest by Professor Bruce C. Berndt of the University of Illinois, in May 1977 and his dedicated efforts for nearly two decades has resulted in the *Ramanujan's Notebooks* published by Springer-Verlag in five Parts, the first of which appeared in 1985. The three original Ramanujan Notebooks are with the Library of the University of Madras, some of the correspondence, papers/letters on or about Ramanujan are with the National Archives at New Delhi and the Tamil Nadu Archives, and a large number of his letters and connected papers/correspondence and notes by Hardy, Watson, Wilson are with the Wren Library of Trinity College, Cambridge. ``Ramanujan : Letters and Commentary'', by Bruce C. Berndt and Robert A. Rankin (published jointly by the American Mathematical Society and London Math. Society, 1995) is a recent publication. The Ramanujan Institute for Advanced Study in Mathematics of the University of Madras is situated at a short distance from the famed Marina Beach and is close to the Administrative Buildings of the University and its Library. The bust of Ramanujan made by Mr. Masilamani is housed in the Ramanujan Institute. In 1992, the Ramanujan Museum was started in the Avvai Kalai Kazhagam in Royapuram. Mrs. Janakiammal Ramanujan, the widow of Ramanujan, lived for several decades in Triplicane, close to the University's Marina Campus and died on April 13, 1994. A bust of Ramanujan, sculpted by Paul Granlund was presented to her and it is now with her adopted son Mr. W. Narayanan, living in Triplicane.